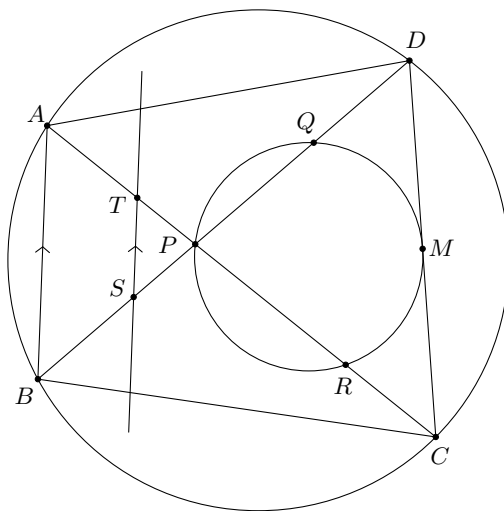


Problem 9. Let P be the intersection point of the diagonals AC and BD in a cyclic quadrilateral $ABCD$. A circle through P touches the side CD in the midpoint M of this side and intersects the segments BD and AC in the points Q and R , respectively. Let S be a point on the segment BD such that $BS = DQ$. The parallel to AB through S intersects AC at T . Prove that $AT = RC$.



Solution. Since $CR \times CP = CM^2 = DM^2 = DQ \times DP$, we have $RC = \frac{DQ \times DP}{CP}$. Now, since $\frac{AT}{DQ} = \frac{AT}{BS} = \frac{AP}{BP}$ we have $AT = \frac{AP \times DQ}{BP}$. The result follows because $ABCD$ is cyclic and so $AP \times CP = BP \times DP$. $\square$

Problem 10. Find all positive integers n such that the decimal representation of n^2 consists of odd digits only.

Solution. The only such numbers are $n = 1$ and $n = 3$.

If n is even, then so is the last digit of n^2 . If n is odd and divisible by 5, then $n = 10k + 5$ for some integer $k \geq 0$ and the second-to-last digit of $n^2 = (10k + 5)^2 = 100k^2 + 100k + 25$ equals 2.

Thus we may restrict ourselves to numbers of the form $n = 10k \pm m$, where $m \in \{1, 3\}$. Then

$$n^2 = (10k \pm m)^2 = 100k^2 \pm 20km + m^2 = 20k(5k \pm m) + m^2$$

and since $m^2 \in \{1, 9\}$, the second-to-last digit of n^2 is even unless the number $20k(5k \pm m)$ is equal to zero. We therefore have $n^2 = m^2$ so $n = 1$ or $n = 3$. These numbers indeed satisfy the required condition. $\square$

Problem 11. The real numbers $x_1, x_2, \dots, x_{2019}$ satisfy

$$x_1 + x_2 = 2y_1, \quad x_2 + x_3 = 2y_2, \quad \dots, \quad x_{2019} + x_1 = 2y_{2019}$$

where $y_1, y_2, \dots, y_{2019}$ is a permutation of $x_1, x_2, \dots, x_{2019}$. Prove that $x_1 = x_2 = \dots = x_{2019}$.

Solution 1. For convenience we call x_{2019} also x_0 . Let k be the largest of the numbers $x_1, x_2, \dots, x_{2019}$, and consider an equation $x_{n-1} + x_n = 2k$, where $1 \leq n \leq 2019$. Hence we get $2 \max\{x_{n-1}, x_n\} \geq x_{n-1} + x_n = 2k$, so either x_{n-1} or x_n , say x_{n-1} , satisfies $x_{n-1} \geq k$. Since also $x_{n-1} \leq k$, we then have $x_{n-1} = k$, and then also $x_n = 2k - x_{n-1} = 2k - k = k$. That is, in such an equation both variables on the left equal k . Now let $\mathcal{E}$ be the set of such equations, and let $\mathcal{S}$ be the set of subscripts on the left of these equations. From $x_n = k$ for any $n \in \mathcal{S}$ we get $|\mathcal{S}| \leq |\mathcal{E}|$. On the other hand, since the total number of appearances of these subscripts is $2|\mathcal{E}|$ and each subscript appears on the left in no more than two equations, we have $2|\mathcal{E}| \leq 2|\mathcal{S}|$. Thus $2|\mathcal{E}| = 2|\mathcal{S}|$, so for each $n \in \mathcal{S}$ the set $\mathcal{E}$ contains both equations with the subscript n on the left. Now assume $1 \in \mathcal{S}$ without loss of generality. Then the equation $x_1 + x_2 = 2k$ belongs to $\mathcal{E}$, so $2 \in \mathcal{S}$. Continuing in this way we find that all subscripts belong to $\mathcal{S}$, so $x_1 = x_2 = \dots = x_{2019} = k$. $\square$

Solution 2. Again we call x_{2019} also x_0 . Taking the square on both sides of all the equations and adding the results, we get

$$\sum_{n=1}^{2019} (x_{n-1} + x_n)^2 = 4 \sum_{n=1}^{2019} y_n^2 = 4 \sum_{n=1}^{2019} x_n^2$$

which can be transformed with some algebra into

$$\sum_{n=1}^{2019} (x_{n-1} - x_n)^2 = 0.$$

Hence the assertion follows. □

Problem 12. There are $2n$ different numbers in a row. By one move we can interchange any two numbers or interchange any three numbers cyclically (choose a, b, c and place a instead of b , b instead of c and c instead of a). What is the minimum number of moves that is always sufficient to arrange the numbers in increasing order?

Solution. If a number y occupies the place where x should be at the end, we draw an arrow $x \rightarrow y$. Clearly at the beginning all numbers are arranged in several cycles: Loops $\bullet \circlearrowleft$, binary cycles $\bullet \rightleftarrows \bullet$ and "long" cycles $\bullet \rightarrow \bullet \rightarrow \bullet \rightarrow \dots$ (at least three numbers). Our aim is to obtain $2n$ loops.



Clearly each binary cycle can be rearranged into two loops by one move. If there is a long cycle with a fragment $\dots \rightarrow a \rightarrow b \rightarrow c \rightarrow \dots$, interchange a, b, c cyclically so that at least two loops, $a \circlearrowleft, b \circlearrowleft$, appear. By each of these moves, the number of loops increase by 2, so at most n moves are needed.

On the other hand, suppose the numbers are $2, 1, 4, 3, \dots, 2n, 2n-1$ in that order. We claim that at least n moves are needed in this case. Note that the configuration at the end is $1, 2, 3, 4, \dots, 2n-1, 2n$. So all the odd numbers must move from even positions to odd positions, while all the even numbers must move from odd positions to even positions. In each move, the parities of the positions of at most two numbers can change. Indeed, this is obvious if we interchange two numbers. If we interchange three numbers a, b, c cyclically, WLOG we may assume both a and b lie in odd positions before the move. Then the parity of the position of a does not change. Hence, at least $\frac{2n}{2} = n$ moves are needed. □